

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

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COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2, BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

CODE:

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Bus Topology Mini Implementation</title>

  <style>

    * {

      box-sizing: border-box;

    }

    body {

      margin: 0;

      font-family: Arial, sans-serif;

      background: #f4f7fb;

      color: #1f2937;

    }

    .container {

      max-width: 1100px;

      margin: 30px auto;

      padding: 20px;

    }

    h1 {

      text-align: center;

      color: #1d4ed8;

    }
```

```
.subtitle {  
  
    text-align: center;  
  
    color: #64748b;  
  
    margin-bottom: 30px;  
  
}  
  
/* Network */  
  
.network {  
  
    background: white;  
  
    padding: 30px;  
  
    border-radius: 15px;  
  
    box-shadow: 0 4px 15px rgba(0,0,0,0.1);  
  
}  
  
.devices {  
  
    display: flex;  
  
    justify-content: space-around;  
  
    gap: 20px;  
  
}  
  
.pc {  
  
    text-align: center;  
  
}  
  
.computer {  
  
    width: 80px;  
  
    height: 55px;  
  
    border: 4px solid #2563eb;  
  
    border-radius: 8px;  
  
    background: #dbeafe;
```

```
        margin: auto;

    }

    .pc-name {

        font-weight: bold;

        margin-top: 15px;

    }

    .ip {

        font-size: 13px;

        color: #64748b;

    }

    /* Common Bus */

    .bus {

        height: 8px;

        background: #334155;

        margin: 40px 60px 15px;

        border-radius: 10px;

    }

    .bus-title {

        text-align: center;

        font-weight: bold;

        color: #475569;

    }

    /* Controls */

    .controls {

        margin-top: 25px;

        display: grid;
```

```
    grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));

    gap: 15px;
}

.card {

    background: white;

    padding: 20px;

    border-radius: 12px;

    box-shadow: 0 3px 10px rgba(0,0,0,0.08);
}

label {

    display: block;

    font-weight: bold;

    margin-bottom: 8px;
}

select,

input,

button {

    width: 100%;

    padding: 11px;

    border-radius: 7px;

    border: 1px solid #cbd5e1;

    font-size: 15px;
}

button {

    background: #2563eb;

    color: white;
```

```
border: none;

cursor: pointer;

font-weight: bold;
}

button:hover {

    background: #1d4ed8;
}

.danger {

    background: #dc2626;
}

.danger:hover {

    background: #b91c1c;
}

.reset {

    background: #64748b;
}

.reset:hover {

    background: #475569;
}

/* Status */

.status {

    margin-top: 25px;

    background: #fff6ff;

    border-left: 5px solid #2563eb;

    padding: 18px;

    border-radius: 8px;
```

```
}

/* Log */

.log {

    margin-top: 20px;

    background: #111827;

    color: #d1fae5;

    padding: 20px;

    border-radius: 10px;

    min-height: 180px;

    font-family: monospace;

    white-space: pre-line;

}

/* Explanation */

.explanation {

    margin-top: 25px;

    background: white;

    padding: 20px;

    border-radius: 12px;

    box-shadow: 0 3px 10px rgba(0,0,0,0.08);

}

.explanation h2 {

    color: #1d4ed8;

}

.explanation li {

    margin-bottom: 8px;

}
```

</style>

</head>

<body>

<div class="container">

<h1>Bus Topology Mini Implementation</h1>

<p class="subtitle">

Small Classroom Network Simulation

</p>

<!-- NETWORK -->

<div class="network">

<div class="devices">

<div class="pc">

<div class="computer"></div>

<div class="pc-name">PC1</div>

<div class="ip">192.168.1.1</div>

</div>

<div class="pc">

<div class="computer"></div>

<div class="pc-name">PC2</div>

<div class="ip">192.168.1.2</div>

</div>

<div class="pc">

<div class="computer"></div>

<div class="pc-name">PC3</div>

<div class="ip">192.168.1.3</div>

</div>


```
<div class="pc">
```

```
  <div class="computer"></div>
```

```
  <div class="pc-name">PC4</div>
```

```
  <div class="ip">192.168.1.4</div>
```

```
</div>
```

```
</div>
```

```
<div class="bus"></div>
```

```
<div class="bus-title">
```

```
  COMMON COMMUNICATION BUS
```

```
</div>
```

```
</div>
```

```
<!-- CONTROLS -->
```

```
<div class="controls">
```

```
  <div class="card">
```

```
    <label>Sender PC</label>
```

```
    <select id="sender">
```

```
      <option value="1">
```

```
        PC1 - 192.168.1.1
```

```
      </option>
```

```
      <option value="2">
```

```
        PC2 - 192.168.1.2
```

```
      </option>
```

```
      <option value="3">
```

```
        PC3 - 192.168.1.3
```

```
      </option>
```

```
      <option value="4">
```

PC4 - 192.168.1.4

</option>

</select>

</div>

<div class="card">

<label>Receiver PC</label>

<select id="receiver">

<option value="2">

PC2 - 192.168.1.2

</option>

<option value="1">

PC1 - 192.168.1.1

</option>

<option value="3">

PC3 - 192.168.1.3

</option>

<option value="4">

PC4 - 192.168.1.4

</option>

</select>

</div>

<div class="card">

<label>Message</label>

<input

type="text"

id="message"

```
        value="Hello from PC1"

    >

</div>

<div class="card">

    <label>&nbsp;</label>

    <button onclick="sendData()">

        Send Data

    </button>

</div>

<div class="card">

    <label>&nbsp;</label>

    <button

        class="danger"

        onclick="busFailure()">

        Simulate Bus Failure

    </button>

</div>

<div class="card">

    <label>&nbsp;</label>

    <button

        class="reset"

        onclick="resetNetwork()">

        Reset Network

    </button>

</div>

</div>
```

<!-- STATUS -->

<div class="status" id="status">

Status:

Network is ready.

Select sender and receiver and send a message.

</div>

<!-- LOG -->

<div class="log" id="log">

SYSTEM LOG

Network initialized.

4 classroom PCs connected.

Common communication bus is active.

Waiting for transmission...

</div>

<!-- EXPLANATION -->

<div class="explanation">

<h2>Mini Implementation Explanation</h2>

<p>

This project demonstrates a small classroom network

using the concept of bus topology.

</p>

Four computers are connected to a common communication bus.

Each computer has a unique IP address.

Data is sent from one PC to another through the common bus.

The system displays whether transmission is successful.

Bus failure can be simulated to demonstrate
the main limitation of bus topology.

</div>

</div>

<script>

// Variable to check whether bus is working

let busWorking = true;

// Send data function

function sendData() {

let sender =

document.getElementById("sender").value;

let receiver =

document.getElementById("receiver").value;

```

let message =

    document.getElementById("message").value;

let status =

    document.getElementById("status");

let log =

    document.getElementById("log");

// Check bus status

if (!busWorking) {

    status.innerHTML =

        "<strong>Status:</strong> " +

        "Transmission failed! " +

        "The common bus is not working.";

    log.innerHTML +=

        "\nBus failure detected." +

        "\nData transmission failed.";

    return;

}

// Sender and receiver cannot be same

if (sender === receiver) {

    status.innerHTML =

        "<strong>Status:</strong> " +

        "Sender and receiver must be different.";

    return;

}

// Check empty message

if (message.trim() === "") {

```

```

status.innerHTML =

    "<strong>Status:</strong> " +

    "Please enter a message.";

return;

}

let senderIP =

    "192.168.1." + sender;

let receiverIP =

    "192.168.1." + receiver;

// Successful transmission

status.innerHTML =

    "<strong>Status:</strong> " +

    "PC" + sender +

    " → COMMON BUS → PC" +

    receiver +

    " | Transmission Successful";

log.innerHTML +=

    "\n\nPC" + sender +

    " (" + senderIP + ")" +

    " is sending: " +

    "\"" + message + "\"" +

    "\nData travels through the common bus..." +

    "\nPC" + receiver +

    " (" + receiverIP + ")" +

    " received the data." +

    "\nTransmission successful!";

```

```

    }

    // Simulate bus failure

    function busFailure() {

        busWorking = false;

        document.getElementById("status").innerHTML =

            "<strong>Status:</strong> " +

            "COMMON BUS FAILURE! " +

            "All communication is disrupted.";

        document.getElementById("log").innerHTML +=

            "\n\nWARNING!" +

            "\nCommon bus cable failure simulated." +

            "\nAll connected PCs are affected.";

    }

    // Reset network

    function resetNetwork() {

        busWorking = true;

        document.getElementById("status").innerHTML =

            "<strong>Status:</strong> " +

            "Network restored. Common bus is active.";

        document.getElementById("log").innerHTML +=

            "\n\nNetwork reset." +

            "\nCommon bus is operational again.";

    }

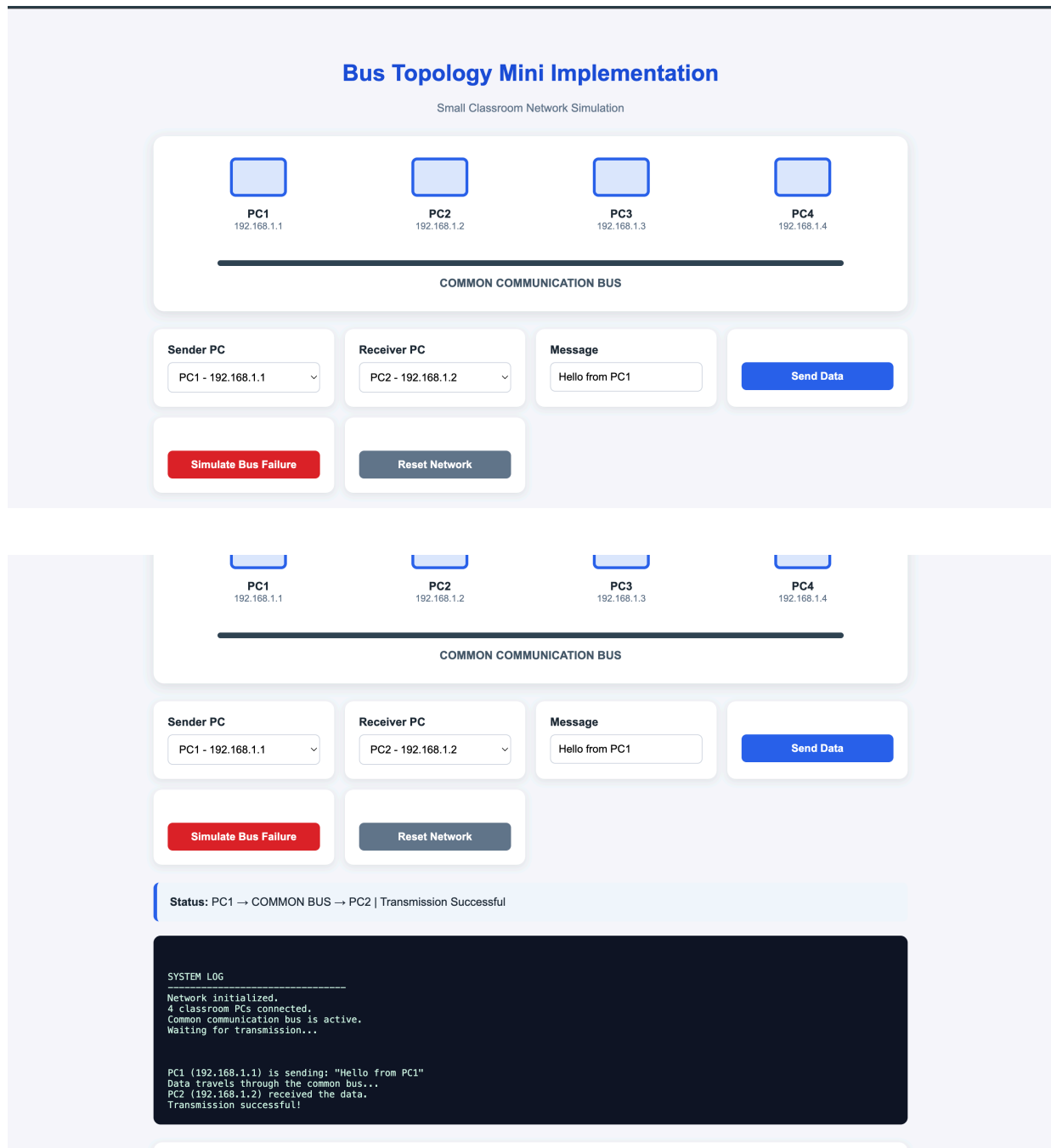
</script>

</body>

</html>

```


OUTPUT:



HOW IT WORKS:

1. The website is created using **HTML, CSS, and JavaScript**.
2. The webpage displays **four computers (PC1, PC2, PC3 and PC4)** connected to a **common communication bus**.
3. Each PC is assigned a unique IP address.
4. The user selects the **Sender PC**, **Receiver PC**, and enters a message.
5. When the **Send Data** button is clicked, JavaScript simulates the data travelling through the common bus to the selected receiver.
6. The **Status** section displays whether the transmission was successful or failed.
7. The **System Log** shows the complete data transmission process.
8. The **Simulate Bus Failure** button demonstrates what happens when the common bus fails.
9. The **Reset Network** button restores the network and allows data transmission again.

Example

PC1 (192.168.1.1)

|

↓

COMMON BUS

|

↓

PC2 (192.168.1.2)

Message: Hello PC2

Output:

Transmission Successful

In this way, the website provides a simple visual simulation of how a bus topology works in a classroom network.

FOR CALCULATION

| Criteria | Weightage | Exemplary (4) | Proficient (3) | Developing (2) | Beginning (1) |
|-----------------------------|-----------|--|---|---|--|
| Problem Analysis | 15% | Clearly identifies and deeply analyzes the industry problem with all factors | Good understanding with minor gaps | Basic understanding; some key aspects missing | Poor or incorrect understanding of the problem |
| Solution Design | 20% | Innovative, practical, and industry-relevant solution | Effective solution with minor limitations | Partially suitable solution | Inappropriate or unclear solution |
| Technical Implementation | 20% | Fully functional, accurate, and well-executed implementation | Mostly functional with minor errors | Partially implemented solution | Incomplete or non-functional implementation |
| Use of Tools & Technologies | 10% | Appropriate and advanced use of relevant tools and technologies | Correct use of tools | Limited or basic use of tools | Incorrect or no use of tools |
| Problem-Solving Approach | 10% | Logical, systematic, and well-structured approach | Mostly logical approach | Somewhat disorganized approach | No clear approach |
| Innovation & Creativity | 10% | Highly creative and original solution | Some creativity | Limited originality | No creativity |
| Testing & Validation | 5% | Thorough testing with real-world scenarios | Adequate testing | Minimal testing | No testing |

| | | | | | |
|------------------------------|----|---|--------------------|-----------------------|----------------------------------|
| Documentation & Presentation | 5% | Clear, professional, and well-documented | Good documentation | Basic documentation | Poor or no documentation |
| Teamwork / Collaboration | 5% | Excellent coordination and contribution (if team-based) | Good collaboration | Limited participation | No collaboration or contribution |